

Mental Health Detection Using Machine Learning: An Enterprise Psychological Risk Screening Framework

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Abstract— Modern enterprise organizations face increasing challenges regarding workplace stress, employee burnout, and reduced productivity. Early psychological intervention and confidential screening are crucial to organizational resilience. This paper presents an AI-powered statistical screening framework using Machine Learning to evaluate demographic, workload, and behavioral risk vectors. Utilizing a tuned Random Forest Classifier (100 estimators, Gini impurity criterion, bootstrap sampling), the model achieves an overall Accuracy of 81.25%, Precision of 0.81, Recall of 0.82, F1-Score of 0.82, and an ROC-AUC of 0.8894. Deployed via a Flask RESTful API with asynchronous JavaScript handling, the system processes submissions in under 200ms and exports timestamped CSV reports. Built-in ethical disclaimers guarantee the tool acts strictly as a preliminary non-diagnostic screening aid.

Index Terms—Mental Health Detection, Machine Learning, Random Forest Classifier, Workplace Wellness, Psychological Risk Screening, Flask REST API.

I. INTRODUCTION

Mental health awareness has become a critical priority in high-pressure corporate and tech work environments. Escalating workplace stress, anxiety, and burnout underscore the need for early identification of employees requiring support. Traditional paper or static survey methods are time-consuming and often fail to detect early-stage risks.

To address this gap, this paper introduces an end-to-end machine learning system that analyzes demographic indicators, workload metrics, family history, and organizational support. By leveraging an ensemble model, the framework computes dynamic probability scores to evaluate psychological strain and support requirements.

A. Organization Context

This study was conducted in collaboration with Decodet Technology (Secunderabad, Telangana), an IT services and consulting firm of 11–50 employees specializing in software development, cloud services, and cybersecurity solutions.

II. PROPOSED RISK FRAMEWORK

Accurate risk assessment requires decomposing psychological strain into multi-dimensional risk vectors across three core pillars:

Fig. 1. Multi-Vector Risk Architecture

- 1. Demographic Vectors:** Role & Seniority, DEI factors, Prior Treatment History.
- 2. Workload Vectors:** Overtime Intensity, Target Realism, Context-Switching.
- 3. External Vectors:** Remote Isolation, Work-Life Boundary Loss, Socio-Economic Strain.

III. SYSTEM METHODOLOGY

The modular ML pipeline executes data cleaning, categorical encoding, scaling, and SMOTE class balancing prior to model inference.

A. Machine Learning Engine

A Random Forest Classifier was chosen for its robustness against overfitting on tabular survey data. Hyperparameters: $n_estimators=100$, $criterion='gini'$, $min_samples_split=2$, $max_features='sqrt'$, and $bootstrap=True$.

IV. RESULTS & DISCUSSION

The Random Forest pipeline achieved superior performance compared to baseline models across all evaluation criteria:

TABLE I. Classifier Performance Comparison

Algorithm	Accuracy	Precision	Recall	F1
Logistic Reg.	74.20%	0.73	0.75	0.74
Decision Tree	72.80%	0.71	0.72	0.71
SVM	78.10%	0.77	0.79	0.78
Random Forest	81.25%	0.81	0.82	0.82

The model achieved an ROC-AUC score of 0.8894, confirming strong class discrimination capability. Integrated into a Flask web platform, inference executes in $\$<200\$$ ms with instant client-side CSV downloads.

V. CONCLUSION & FUTURE WORK

The framework successfully bridges ML analytics and workplace wellness screening, delivering high-accuracy risk evaluations alongside non-diagnostic ethical safeguards.

A. Demographic Vectors

Junior employees face role instability and fear of speaking up; mid-level managers suffer "sandwich pressure" balancing execution and emotional care; executives face high-stakes isolation.

B. Workload & External Vectors

Chronic overtime degrades cognitive function, while aggressive targets maintain elevated cortisol levels. Remote work exacerbates social detachment and boundary blur.

Future work includes Privacy-Preserving ML (Federated Learning) and Zero Trust RBAC architecture.

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